Your AI Needs a Boss; Appoint an AI Leader

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Initiatives:Artificial Intelligence; Establish a World-Class AI Strategy and Organization

In today's world, organizations without an AI leader will struggle to survive. Realizing the full potential of AI-driven transformation across business units and functions requires an AI leader — a transformative role that blends business acumen, strategic insight and technological expertise to effectively balance business value with AI technological innovation throughout the organization.

More on This Topic

This is part of an in-depth collection of research. See the collection:

- Leaders Who Fail to Embed AI Into Their Enterprise's DNA Will Be Rapidly Outpaced

Quick Answer

What is an AI leader?

- An AI leader is the seniormost executive in the organization in charge of AI. AI leaders wield knowledge, AI expertise and IT skills. They should possess three key competencies: business and strategic knowledge; technology knowledge; and communication skills.
- AI leaders' reporting relationships depend on the scope of their role. Which role they report to will depend on a mixture of core competencies and the core mission of the organization.
- AI leaders manifest their organization's AI ambitions. The focus will be shaped by organizational context and AI ambition.

More Detail

In today's world, organizations without an AI leader will struggle to survive.

Appointing a dedicated AI leader is key for an organization's AI maturity and success. This is clearly demonstrated in our 2024 Gartner AI Mandates for the Enterprise Survey. Only 37% of low-maturity organizations report having dedicated AI leaders, compared to an impressive 91% of high-maturity organizations.

Among organizations with a dedicated AI leader, the survey also found that:

- 38% have their AI projects stay in production for more than three years, compared to less than 21% of organizations that don't have an AI leader.
- 42% have their business's trust and ready-to-use technology, compared to less than 19% of organizations that don't have an AI leader.
- Strong performance in key business outcomes is twice as likely compared to those without an AI leader: generating revenue (28% vs. 13%), reducing costs (28% vs. 15%) and excelling in customer experience (28% vs. 17%).
- 44% of generative AI (GenAI) prototypes make it to production, compared to less than 36% in organizations that don't have an AI leader.

The AI leader is a stand-alone role. Organizations with high AI maturity recognize that assigning AI leadership responsibilities to existing roles can create conflicts of interest — such as balancing the demands of maintaining and evolving current business operations with the need to drive AI-enabled processes and innovation.

Regardless of the path an organization chooses, failing to establish clear AI leadership increases the risk of falling behind competitors that have defined and empowered this critical role. Also, in the public sector, political and societal stakeholders are increasingly pressing organizations to leverage AI to improve responsiveness, reduce costs or achieve other benefits.

What Is an AI Leader?

While organizations have had dedicated AI teams for years, there was often a gap: effective AI leadership. That void is now being filled, with organizations increasingly establishing the role of AI leader.

Most organizations start with a partial AI leader — that is, an existing leader (often a CIO or CTO) with additional AI responsibilities. But as the organization matures, it becomes apparent that successful AI really requires a dedicated position responsible and accountable for the delivery and scaling of AI value throughout the organization.

The AI leader is the seniormost executive in the organization in charge of AI.

AI leaders often have a broad scope and mandate for AI-enabled change, spanning both the technical and business dimensions of the enterprise.

In 2024, a proliferation of advanced AI solutions within organizations increased the proportion of AI teams reporting to CTOs and dedicated AI leaders. This was due to the increased use of the technology with applications and the emerging technology assessment duties that these leaders often have.

The wide applicability of AI and its business-transformative nature means that CIOs and CTOs can no longer take on direct responsibility for AI. The dedicated AI leader role thus becomes necessary to take on both the business and technology leadership of AI in the organization. The reporting lines of AI leaders will depend on the company's AI ambition, which is often shaped by company or industry core competencies.

AI leader roles can often begin informally in organizations as an extra duty on top of existing staff responsibilities. However, this is not a sustainable approach, given:

- The growth in demand for AI
- The scope of AI reach within business units and functions
- The rate at which the technology is evolving and impacting business

The impact and potential of AI in organizations is transformative and requires dedicated leadership and resources.

Today, we see two broad classes of dedicated AI leadership roles:

1. **Leadership roles solely dedicated to AI.** These include job titles such as head of AI, chief AI officer, VP of AI, and director of AI.

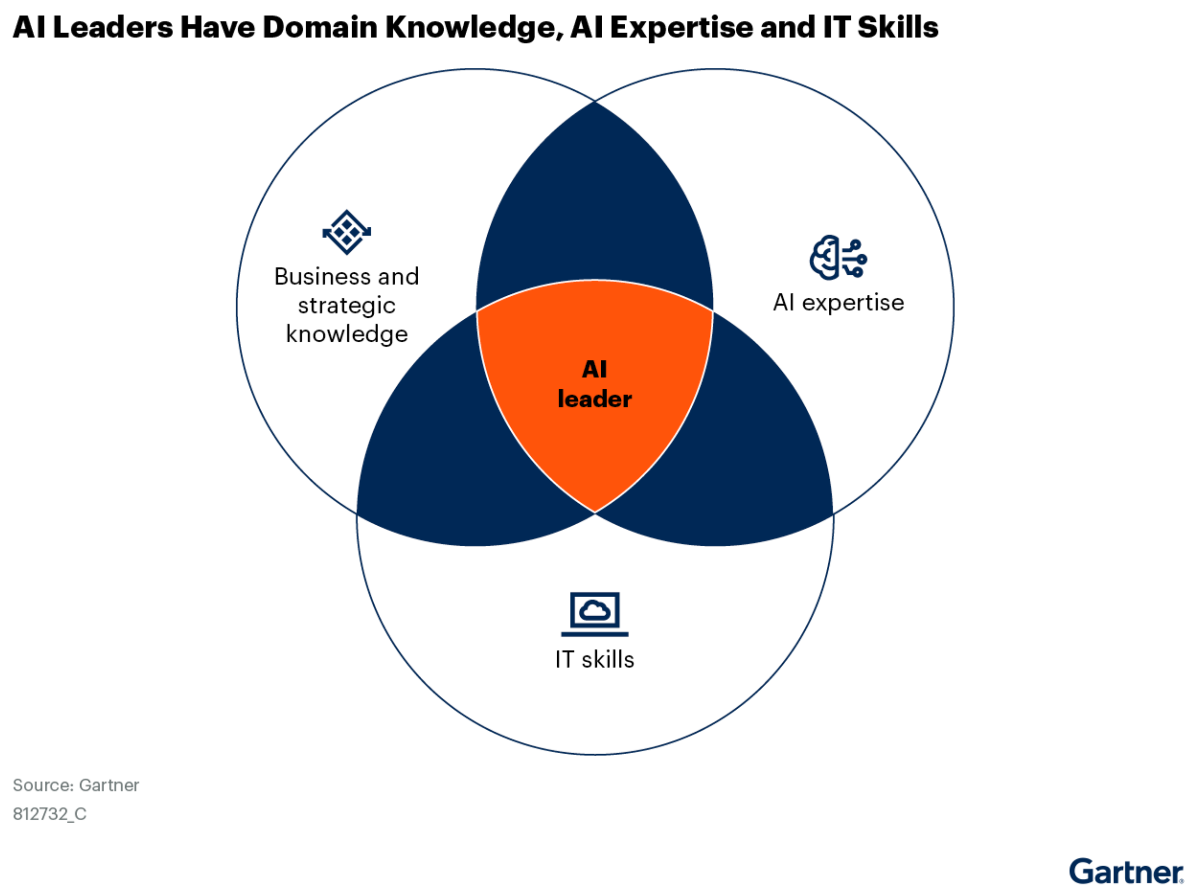
2. **Roles with AI as an additional responsibility.** These roles are many and varied. Titles include global head of data and AI, head of digital transformation and AI, VP of operations and AI, head of AI and automation transformation, chief data and analytics and AI officer, and so on.

The AI leader is not a part-time role. The impact and potential of AI in organizations is transformative and requires dedicated leadership.

AI Leaders Wield Business and Strategic Knowledge, AI Expertise, and IT Skills

AI leaders sit at the intersection of AI expertise, IT skills and business/strategic knowledge (see Figure 1). Finding the right candidate can be a challenge due to the scope of the role, which requires experience in multiple AI projects and systems, from design and business modeling to detailed development, operations and change management. A chief data scientist with little IT or business knowledge, for example, would not be suitable as an AI leader in the same way that a software engineering leader with no AI expertise wouldn't be appropriate. "AI" in this case refers to the full spectrum of AI techniques (i.e., machine learning as well as symbolic techniques) and AI practices (such as generative AI, agentic AI, decision intelligence, data science and composite AI).

Figure 1: AI Leaders Have Domain Knowledge, AI Expertise and IT Skills



Three Key Skills

AI leaders need to have three key competencies:

1. Enough business and strategic knowledge to align with the highest levels of the organization, and to communicate complex issues and resolutions to the board.

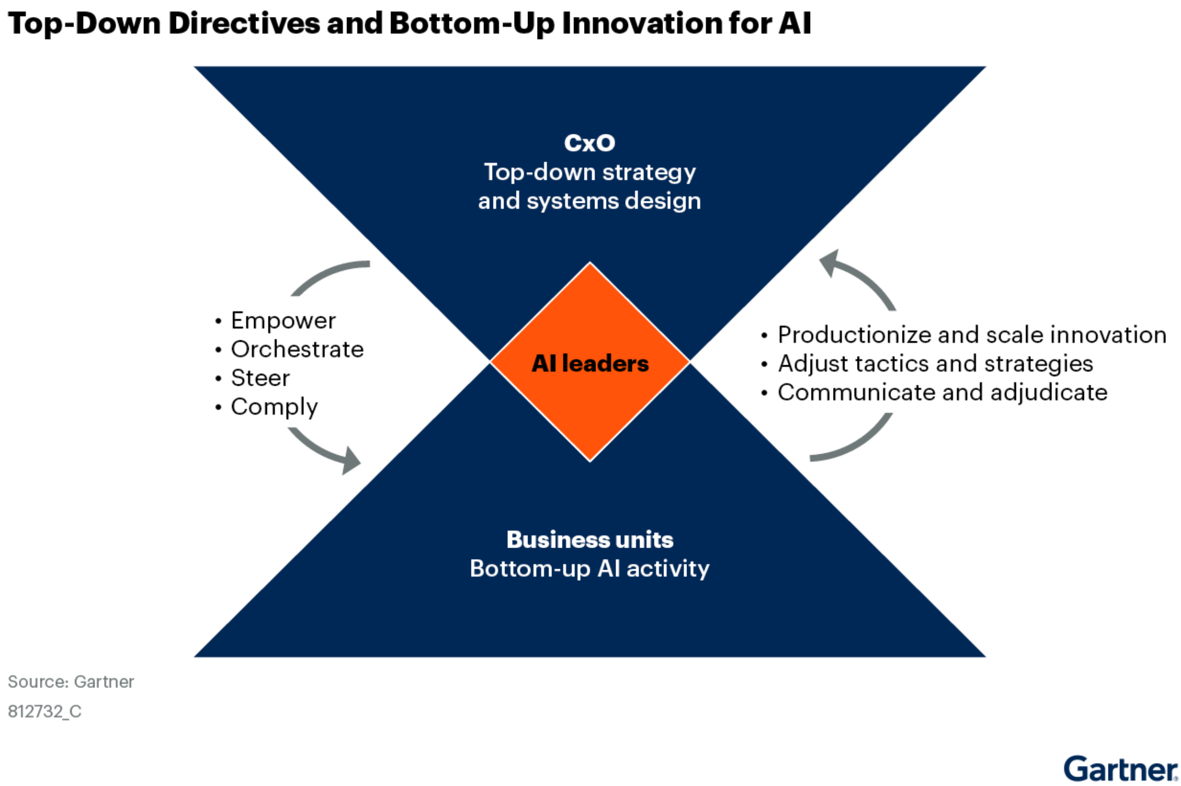
2. Enough technology knowledge and AI expertise to understand how to apply and orchestrate AI throughout the organization and, as a result, gain trust from business units and IT as collaborators.

3. Communication skills to build bridges across organizational layers and across business units and functions to manage expectations and mitigate fears brought by AI deployments.

While AI leaders may come from an existing AI, scientific or IT role, they transcend these individual practices. For instance, they can set strategy as well as guide the design of AI systems spanning enterprise business and technology capabilities.

AI leaders can't — and shouldn't — control all aspects of AI in an organization. To thrive in AI's transformative landscape, organizations require an AI leader who marries executive strategy with operational agility. Figure 2 illustrates this balance, highlighting the AI leader's role in harmonizing executive direction with the innovative pulse of business units.

Figure 2: Top-Down Directives and Bottom-Up Innovation for AI



The superset task for AI leaders is to be the linchpin between the C-suite's top-down strategies and system design directives, and the bottom-up AI activities from various business units (BU heads).

AI Leaders Need to Ramp Up Quickly

AI leaders, when taking a leadership role inside the organization, should quickly ramp up by:

- **Clarifying the C-suite's view of AI leadership.** Understand the role of AI leadership through the lens of current executives and align it with the organization's business and digital imperatives.

- **Refreshing their own industry knowledge.** Enhance their knowledge of the industry, competitors and strategic factors reliant on AI for superior execution.
- **Getting a clear mandate from the executive committee.** Define clear expectations with C-suite colleagues regarding their AI mandate, priorities and decision-making authority (see Table 1). Ask for the first wave of dedicated resources.
- **Assessing the organization's AI maturity.** Identify existing AI projects and competencies. From there, evaluate the AI readiness and aspirations of both leadership and the organization to enable adept navigation of the technology transformation.

Along with these immediate tasks to frame their mandate, a mixture of ongoing and one-off tasks will help determine the vision and the AI readiness of the organization. While the tasks are ultimately shaped by an organization's ambition for AI and the positioning of AI leaders within the organization, all AI leaders have a common set of tasks. These are covered in the next section, and further on in Tables 2 through 5.

AI Leaders Manifest Their Organization's AI Ambitions

While the focus and operating model for AI leaders will be shaped by organizational context and AI ambition, the common threads and responsibilities are to:

- **Accelerate AI-driven transformation.** AI leaders must build and/or maintain a vision of where and how AI will be used in the organization to drive value aligned to strategic objectives. This should include how AI will impact current and future business objectives, going beyond financial metrics to cover differentiating ideas, employee value propositions and other factors. Getting executive and business unit buy-in to a single vision of benefit allows investments to be prioritized and payoff decisions to be made effectively. It creates a single view that the wider enterprise can be held accountable for in moving forward with AI. AI leaders should also be able to articulate the transformational impact that AI could have on their organization and potentially their industry.

- **Establish a world-class AI strategy and organization.** AI leaders play a crucial role in executing a world-class AI strategy that aligns with the company's overarching vision. They also play a role in evolving operating models to match that vision. This includes goal setting and measuring as well as incorporating research partnerships and value models into the strategic framework. They must decide when AI is worth the cost and make the value from AI measurable, transparent and attributable. AI leaders should have organizational and structural responsibilities, and assess whether their organization needs centralized or decentralized AI competencies. They are instrumental in developing business models, assembling competent teams, and establishing frameworks for assessing value, risk and policy development. Following these steps will ensure the organization's AI initiatives drive significant and sustainable impact.

- **Build, engineer and implement AI initiatives.** An AI leader is responsible for spearheading the development and execution of AI initiatives. This includes both conceptualizing and engineering advanced AI systems and solutions and refining delivery models for optimal efficiency. They also ensure the scalability and seamless integration of AI solutions across the organization. This includes orchestrating the deployment of AI technologies to drive transformative outcomes while establishing observability mechanisms that maintain the performance and ethical implementation of AI techniques.

- **Develop cross-organizational and functional alignment.** An AI leader plays a pivotal role in ensuring cross-organizational alignment, responsible AI implementation and stakeholder engagement. They play a role in defining collaboration frameworks and orchestrating AI technologies to meet strategic objectives and drive innovation. AI leaders bridge departmental gaps, mitigate risks, and manage AI platforms to enhance business strategy and operational efficiency. This will ensure that AI initiatives deliver maximum value while limiting redundancies, avoiding potential conflicting implementations, and adhering to robust AI governance frameworks that establish best practices and compliance standards.

- **Select and orchestrate AI platforms and technologies.** An AI leader is charged with the selection and coordination of AI platforms and AI embedded in enterprise applications, and ensuring alignment with the technical needs and pre-existing investments of the organization. Furthermore, AI leaders are responsible for staying abreast of market and technology trends and emerging AI innovations, which will enable them to set strategy and maintain the organization's competitive edge through the adoption of cutting-edge AI solutions. Finally, they must help build and maintain novel data management methods, new processing units and computing environments for training and execution of AI systems.

The appointment of a formal AI leader acts as a catalyst for enterprise operating models to evolve and adapt. Without the role, operating models can often stall and languish in suboptimal best-effort approaches, which can result in enterprises getting stuck in the "productivity trap." Table 1 outlines a typical (not definitive) dynamic for AI leaders in relation to other organizational practices in the form of a responsible, accountable, consulted and informed (RACI) model.

Table 1: Broad RACI Model for AI and Other Leaders Straddling AI Enterprise Technologies

(Enlarged table in Appendix)

Activities	AI leader	EA	SWE	I&O	D&A	Business unit	Apps	HR/legal
AI culture	A, R	I	C	C	C	R	C	R
AI governance	R	A	R	R	R	R	R	C
AI research	R, A	C	C	I	R	I	I	I
AI strategy	A, R	R	C	C	C	C	C	C
AI design	A, R	R, C	R	C	R	I	R	C
AI business cases	R	R	C	C	C	A	C	C
Buy AI solutions	A	R	C	C	C	R	C	C
Develop AI solutions	A	R, C	R	C	R	R	R	C
Test AI solutions	A	C	R	C	R	R	R	C
Operate AI	R	C	C	A	R	R	R	C
Realize AI benefits	R	C	C	C	C	A	C	C

R = responsible — responsible for shaping and executing the task; A = accountable — delegates work and final review (only one person can be accountable); C = consulted — provides input on impacts along with domain expertise; I = informed — kept in the loop.
EA = enterprise architecture; SWE = software engineering; I&O = infrastructure and operations; D&A = data and analytics; apps = applications; HR = human resources.

Source: Gartner

AI Leaders' Reporting Relationship Depends on the Scope of Their Role

Who the new AI leader reports to will depend on a mixture of the core competencies and the core mission of the organization, and where it sees AI as most critical to supporting that mission.

Organizations must answer two questions:

1. **Where can AI provide the greatest impact and influence on enterprise missions?** To ensure that the organization's AI strategy and goals are effectively achieved, it is crucial for AI leaders to report to the right enterprise, division or department, at a level where they can have the greatest impact and exert the most influence.

2. **Where can AI leaders achieve seamless organizational coordination and alignment?** The reporting structure should ensure the integration of the AI initiatives with the overall organization's objectives.

AI leaders do not only report to technology roles like CIO and CTO. Today, we see many examples of AI leaders reporting to:

- A chief medical officer in the health industry to focus on patient care and clinical decision making.
- A chief risk officer in the financial services industry.
- A chief science officer in an R&D organization to tie AI efforts to the organization's scientific vision.
- A chief operations officer in the manufacturing industry to optimize production processes.
- A chief product officer for a retail organization.
- A chief executive officer in a pharmaceutical company that considers AI as a pivotal game changer throughout the entire organization.

Tables 2 through 5 detail the superset of tasks that underpin the primary goals of the AI leader in any organization.

Table 2: Establish a World-Class AI Strategy and Organization

High-level task	Definition
Establish AI use cases and business frameworks	Identify a portfolio of potential AI use cases and an initial roadmap based on feasibility, AI maturity, organizational readiness and potential benefits.
Develop AI vision, strategy and innovation	Build an AI-first strategy and vision while maintaining board and senior leadership communications and managing stakeholders' involvement.
Build a world-class AI organization and team	Set up an effective AI organization, define new roles and set up a talent ecosystem while managing expectations across the organization.
AI-enabled business transformation and new business models	Assess and communicate the transformative impact of AI through new business models and the regeneration of existing workflows.
Assess risk to develop safe and secure AI policies	Identify, assess, measure and monitor business and technology risks. Build and enforce AI policies across the organization in collaboration with the CISO.
COE = center of excellence; KPI = key performance indicator.	

Source: Gartner

Table 3: Build, Engineer and Implement AI Initiatives

High-Level Task	Definition
Design and engineer AI systems and solutions	Establish practices to design, engineer and simulate AI systems to embed AI techniques within applications, business processes or digital twins across the organization.
Develop and improve AI delivery models	Institute flexible and robust AI delivery models through a unified AI governance framework and a product-focused management model, thus curbing shadow AI.
Operationalize and safely manage AI solutions at scale	Develop robust, flexible, secure and sustainable frameworks for the operationalization and orchestration of AI solutions at scale.
Estimate, measure and track AI value and ROI	Identify business, technical and operational metrics while building, monitoring and measuring the ongoing impact of use cases and their ripple effects.

Source: Gartner

Table 4: Develop Cross-Organizational and Functional Alignment

High-Level Task	Definition
Build synergy with business units and functions	Establish collaboration networks and councils across the organization through fusion teams to monitor and implement AI solutions.
Align AI initiatives across technology stakeholders	Align AI seamlessly across workflows and processes, including IT operations such as D&A, apps management, software engineering, IT, and risk and security.
Foster AI governance through compliant and responsible AI	Determine the shape and reach of AI governance boards to guarantee compliance with AI regulations associated with AI systems.
Develop change management and AI literacy programs	Establish AI-induced change management and AI communication programs while educating the organization, from the board to developers.

Source: Gartner

Table 5: Select and Orchestrate AI Platforms and Technologies

High-Level Task	Definition
Track AI markets and technology trends	Track market evolutions for new technologies and trends while assessing how (and whether) new AI models can benefit the organization.
Buy and coordinate AI platforms and technologies	Coordinate the management and selection of cross-organizational AI platforms while establishing best practices and effective technology reuse across the organization.
Create and maintain a data and AI infrastructure	Build and maintain novel data management methods, new processing units (e.g., GPUs) and computing environments for training and execution of AI systems.
GPU = graphical processing unit	

Source: Gartner

Evidence

Social Media Analytics Team. Gartner conducts social listening analysis using third-party data tools (leveraging LinkedIn, O*NET and other job boards) to complement or supplement the other fact bases presented in this document. Due to their qualitative and organic nature, the results should not be used separately from the rest of this research. No conclusions should be drawn from this data alone. Social media data referenced is for January 2024 in all geographies (except China) and recognized languages.

[1] **2024 Gartner AI Mandates for the Enterprise Survey.** This study was conducted to understand how AI and generative AI (GenAI) are being adopted by enterprises, focusing on areas such as AI strategy, data, governance, literacy, engineering, organization, portfolio and value, to assist clients in keeping pace with AI's rapid evolution. The research was conducted online from October through December 2024 among 432 respondents from the U.S. (n = 181), the U.K. (n = 70), France (n = 50), Germany (n = 50), India (n = 51) and Japan (n = 30). Quotas were established for company sizes and for industries to ensure a good representation across the sample. Organizations were required to have deployed at least one AI use case in production. Respondents were screened for C-level executives (e.g., chief AI officer, chief data officer, chief data scientist, chief digital officer, chief information officer, chief operating officer, chief technology officer or equivalent) or roles above vice presidents. All respondents were required to have high involvement in at least one AI initiative. Disclaimer: The results of this survey do not represent global findings or the market as a whole, but reflect the sentiments of the respondents and companies surveyed.

Document Revision History

Quick Answer: What Is an AI Leader? - 26 April 2024

Recommended by the Authors

Some documents may not be available as part of your current Gartner subscription.

AI Leaders' Roles and Skills, Worldwide, 2024

Gartner AI Maturity Model

Toolkit: Build Your Organization's AI Roadmap

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